Python Basics - Assingment 5

September 30, 2021

1 Exercise 5

Task 1. Write a program that prompts students to provide their total points (0 -100) and check with the following conditions:

- if total point is greater than or equal to 85 then the grade is 5
- if total point is greater than or equal to 75 then the grade is 4
- if total point is greater than or equal to 65 then the grade is 3
- if total point is greater than or equal to 55 then the grade is 2
- if total point is greater than or equal to 45 then the grade is 1
- if total point is less than 45 then the grade is 0

(Use IF statement)

[2]:
```
total_points = 100

if total_points >= 85:
  print("the grade is 5")
elif total_points >= 75:
  print("the grade is 4")
elif total_points >= 65:
  print("the grade is 3")
elif total_points >= 55:
  print("the grade is 2")
elif total_points >= 45:
  print("the grade is 1")
elif total_points < 45:
  print("the grade is 0")
```

```
the grade is 5
```

1.0.1 Example solutions Task 1.

CLICK ME

```
#ask user to input point (Hint: use int() to cast the input to integer)
# write conditions to check and output the grade based on point


# Example
```

```
points = int(input("Points:"))
if points >= 85:
    print ("Grade: 5")
elif points >= 75:
    print ("Grade: 4")
elif points >= 65:
    print ("Grade: 3")
elif points >= 55:
    print ("Grade: 2")
elif points >= 45:
    print ("Grade: 1")
else:
    print ("Grade: 0")
```

Task 2. Write a program that asks user to input a number and create the multiplication table (from 1 to 10) of a number (Use For). The output should be as below for an inputted number:

```
5 * 1 = 5
5 * 2 = 10
...
...
...
5 * 10 = 50
```

[9]:
```
number = int(input("Enter a number:"))
for x in range(1,11):
    print(number ," * ", x , "= " ,(number * x))
```

```
Enter a number: 5

5  *  1 =  5
5  *  2 =  10
5  *  3 =  15
5  *  4 =  20
5  *  5 =  25
5  *  6 =  30
5  *  7 =  35
5  *  8 =  40
5  *  9 =  45
5  *  10 =  50
```

1.0.2 Example solutions Task 2.

CLICK ME

```
# Example
num = int(input("Enter a number:"))
for i in range(1, 11):
```

```
    print (str(num) + " * " + str(i) + " = " + str(num * i))
```

Task 3. Write a program using Numpy to generate and print 5 random number between 0 & 1.

[12]:
```
import numpy as np
from numpy import random
for i in range(5):
    print (random.rand())
```

```
0.49892068560473146
0.6346675614498514
0.6884426869144114
0.13326727134140948
0.7191276380580977
```

1.0.3 Example solutions Task 3.

CLICK ME

```
 # print the 5 random numbers

# Example
import numpy as np

for i in range(5):
    print (np.random.random())
```

Task 4. Write a function to print even numbers from a list.

[13]:
```
list = [12,24,33,44,15,36,17,78,97,107]
def even(listOfNumber):
    for x in list:
        if x % 2 == 0:
            print(x)


even(list)
```

```
12
24
44
36
78
```

1.0.4 Example solutions Task 4.

CLICK ME

```
# create an integer list that has 10 numbers 1,2,3,4,5,6,7,8,9,10
# check and print all even numbers from the list (use a function)

# Example
def printEven(list):
    for n in list:
        if n % 2 == 0:
            print (n)

mylist = [1,2,3,4,5,6,7,8,9,10]
printEven(mylist)
```

Task 5. Write a python program to write the courses list into a file called mycourses.txt and print the content of the file.

[17]:
```
courses = ['English','Math','Algebra','science','History']
f = open("mycourses.txt", "x")
f = open("mycourses.txt", "w")
f.write(str(courses))
f = open("mycourses.txt", "r")

print(f.read())
f.close()
```

```
['English', 'Math', 'Algebra', 'science', 'History']
```

1.0.5 Example solutions Task 5.

CLICK ME

```
#use the list below
courses = ['Introduction to IoT', 'Basics of Python', 'Python Project']

# Example
f = open("mycourses.txt", "w")
f.write(str(courses))
f.close()
f = open("mycourses.txt", "r")
print (f.read())
f.close()
```

Task 6. Append the file created in no. 5 and add the text "Here is more text".

[18]:
```
f = open("mycourses.txt", "a")
f.write("Here is more text")
f.close()
```

1.0.6 Example solutions Task 6.

CLICK ME

```
# Example
f = open("mycourses.txt", "a")
f.write("Here is more text")
f.close
```

[ ]:
```

```